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Chapter 3.

$\mathbb{H}^A \otimes \mathbb{H}^B$.

ONB $\{|e_a \otimes f_b\rangle \mid \{|e_a\rangle\}$: ONB of $\mathbb{H}^A$, $\{|f_b\rangle\}$: ONB of $\mathbb{H}^B\}$

$$|\Psi\rangle = \sum_{a,b} \Psi_{ab} |e_a \otimes f_b\rangle$$

inner product $\langle \Psi | \Phi \rangle = \sum_{a,b} \overline{\Psi_{ab}} \Phi_{ab}$.

→ positive definite

$(\because \ |\Psi\rangle = \sum_{a,b} \Psi_{ab} |e_a \otimes f_b\rangle$

$\Rightarrow \langle \Psi | \Psi \rangle = \sum_{a,b} |\Psi_{ab}|^2 \geq 0$

$\langle \Psi | \Psi \rangle = 0$ iff $\forall \Psi_{ab} = 0$ iff $\Psi = 0$ $)$

→ × affected by the choice of ONB.

(Fact : $|\hat{e}_a\rangle = U^A |e_a\rangle$: ONB of Hilbert sp $\Leftrightarrow$ U : Unitary operator).

Induced norm $\|\Psi\| = \sqrt{\langle \Psi | \Psi \rangle} = \sqrt{\sum_{a,b} |\Psi_{ab}|^2}$.

→ $\|\varphi \otimes \Psi\| = \sqrt{\langle \varphi \otimes \Psi | \varphi \otimes \Psi \rangle} = \sqrt{\langle \varphi | \varphi \rangle \langle \Psi | \Psi \rangle} = \|\varphi\| \|\Psi\|$

for $|\varphi\rangle \in \mathbb{H}^A$, $|\Psi\rangle \in \mathbb{H}^B$.

Summary.

$\mathbb{H}^A \otimes \mathbb{H}^B$: Hilbert sp w. "tensor product of $\mathbb{H}^A$ & $\mathbb{H}^B$"

$$\begin{cases} \text{inner prod} & \langle \Psi | \Phi \rangle = \sum_{a,b} \overline{\Psi_{ab}} \, \Phi_{ab} \\ \text{norm} & \| \Psi \| = \sqrt{\sum_{a,b} |\Psi_{ab}|^2} \\ \text{ONB} & \{ |e_a \otimes f_b\rangle \mid \{|e_a\rangle\} : \text{ONB of } \mathbb{H}^A, \{|f_b\rangle\} : \text{ONB of } \mathbb{H}^B \} \\ \dim(\mathbb{H}^A \otimes \mathbb{H}^B) & = \dim \mathbb{H}^A \cdot \dim \mathbb{H}^B \quad (\text{for finite-dim } \mathbb{H}^A, \mathbb{H}^B) \end{cases}$$

cf. $\{|e_a\rangle\}$: basis of $\mathbb{H}^A$

$\{|f_b\rangle\}$: " $\mathbb{H}^B$

$\{|g_c\rangle\}$: " $\mathbb{H}^C$

$$\Rightarrow (\mathbb{H}^A \otimes \mathbb{H}^B) \otimes \mathbb{H}^C = \mathbb{H}^A \otimes (\mathbb{H}^B \otimes \mathbb{H}^C)$$

$$= \mathbb{H}^A \otimes \mathbb{H}^B \otimes \mathbb{H}^C.$$

$$\langle \varphi_1 \otimes \psi_1 \otimes \chi_1 | \varphi_2 \otimes \psi_2 \otimes \chi_2 \rangle = \langle \varphi_1 | \varphi_2 \rangle \langle \psi_1 | \psi_2 \rangle \langle \chi_1 | \chi_2 \rangle$$

$$|\Psi\rangle \in \mathbb{H}^A \otimes \mathbb{H}^B \otimes \mathbb{H}^C \rightarrow |\Psi\rangle = \sum_{a,b,c} \Psi_{abc} |e_a \otimes f_b \otimes g_c\rangle$$

$$\mathbb{H}^A \simeq \mathbb{C}^{n_A}$$

$$|e_1\rangle \mapsto \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

$$(\mathbb{H}^A)^* \simeq \mathbb{H}^A \simeq \mathbb{C}^{n_A}, \quad \mathbb{H}^B \simeq \mathbb{C}^{n_B} \Rightarrow \mathbb{H}^A \otimes \mathbb{H}^B \simeq \mathbb{C}^{n_A n_B}$$

$$|e_1 \otimes f_1\rangle = \begin{array}{cc} \begin{pmatrix} 1 \\ 0 \\ \vdots \\ \vdots \\ 0 \end{pmatrix} & \begin{array}{l} \leftarrow 1 \\ \\ \\ \\ \leftarrow n_A n_B \end{array} \end{array} \quad |e_1 \otimes f_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad \cdots, \quad |e_a \otimes f_b\rangle = \begin{array}{cc} \begin{array}{r} \\ \\ (a-1)n_B + b \rightarrow \\ \\ \\ \end{array} & \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \end{array}$$

$$\langle e_a \otimes f_b| = (0 \cdots \underset{\substack{\uparrow \\ (a-1)n_B + b}}{1} \cdots 0).$$

ex) $\mathbb{H}^A = \mathbb{H}^B \simeq \mathbb{C}^2$

ONB of $\mathbb{H}^A, \mathbb{H}^B$: $\{|e_a\rangle\} = \{|f_b\rangle\} = \{|0\rangle, |1\rangle\} = \left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\}.$

$\Rightarrow$ ONB of $\mathbb{H}^A \otimes \mathbb{H}^B \simeq \mathbb{C}^4$: $\{|e_a \otimes f_b\rangle\} = \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$

$$= \left\{\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}\right\}.$$

$$\begin{array}{ccc} \mathbb{H}^A & & \mathbb{H}^B \\ \cup\!\!\!\mid & & \cup\!\!\!\mid \end{array}$$

$$\Rightarrow \quad |\varphi\rangle = a_1|0\rangle + b_1|1\rangle, \qquad |\Psi\rangle = a_2|0\rangle + b_2|1\rangle,$$

$$|\varphi\otimes\Psi\rangle = (a_1|0\rangle + b_1|1\rangle)\otimes(a_2|0\rangle + b_2|1\rangle)$$

$$= a_1a_2|00\rangle + a_1b_2|01\rangle + b_1a_2|10\rangle + b_1b_2|11\rangle$$

$$= \begin{pmatrix} a_1a_2 \\ a_1b_2 \\ b_1a_2 \\ b_1b_2 \end{pmatrix}.$$

$$\langle\varphi\otimes\Psi| = (a_1a_2 \quad a_1b_2 \quad b_1a_2 \quad b_1b_2)$$

Definition

- Tensor product of qubit spaces

$$^q\mathbb{H}^{\otimes n} := \underbrace{^q\mathbb{H}\otimes\cdots\otimes{}^q\mathbb{H}}_{n \text{ times}} \qquad \rightarrow \dim {}^q\mathbb{H}^{\otimes n} : 2^n$$

- $(j+1)$-th factor space $^q\mathbb{H}_j$ counts from the right!

$$^q\mathbb{H}^{\otimes n} = {}^q\mathbb{H}_{n-1}\otimes{}^q\mathbb{H}_{n-2}\otimes\cdots\otimes\underbrace{^q\mathbb{H}_j}_{(j+1)\text{th}}\otimes\cdots\otimes{}^q\mathbb{H}_0$$

* Basis of ${}^q\mathbb{H}^{\otimes n}$

1. Computational basis.

For $x \in \mathbb{N}_0$, $x < 2^n$,

$x = x_{n-1} x_{n-2} \cdots x_1 x_0 \,_{(2)}$ $\quad (x_i \in \{0,1\})$

Define $|x\rangle^n := |x\rangle := |x_{n-1} x_{n-2} \cdots x_1 x_0\rangle := |x_{n-1}\rangle \otimes |x_{n-2}\rangle \otimes \cdots \otimes |x_0\rangle.$

$$= \bigotimes_{j=n-1}^{0} |x_j\rangle$$

$\rightarrow$ ${}^q\mathbb{H}^{\otimes n} = {}^q\mathbb{H}_{n-1} \otimes \cdots \otimes {}^q\mathbb{H}_j \otimes \cdots \otimes {}^q\mathbb{H}_0$

$\ni |x_{n-1}\rangle \otimes \cdots \otimes |x_j\rangle \otimes \cdots \otimes |x_0\rangle$

<u>lemma 3.7</u>

$\{ |x\rangle \in {}^q\mathbb{H}^{\otimes n} \mid x \in \mathbb{N}_0 , x < 2^n \}$: ONB of ${}^q\mathbb{H}^{\otimes n}$.

Pf) 1) $\langle x | y \rangle = \langle x_{n-1} x_{n-2} \cdots x_1 x_0 | y_{n-1} y_{n-2} \cdots y_1 y_0 \rangle$

$= \langle x_{n-1} | y_{n-1} \rangle \langle x_{n-2} | y_{n-2} \rangle \cdots \langle x_0 | y_0 \rangle$

$= \delta_{xy}$ $\quad \therefore$ orthonormal

2) $| \{ |x\rangle \} | = 2^n = \dim {}^q\mathbb{H}^{\otimes n}.$

ex) In ${}^1\mathbb{H}$ $\quad |0\rangle^1 = |0\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \quad |1\rangle^1 = |1\rangle = \begin{pmatrix}0\\1\end{pmatrix}.$

In ${}^1\mathbb{H}^{\otimes 2} \simeq \mathbb{C}^4$ $\quad |0\rangle^2 = |00\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \quad |1\rangle^2 = |01\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix},$

$$|2\rangle^2 = |10\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \quad |3\rangle^2 = |11\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}.$$

In ${}^1\mathbb{H}^{\otimes 3} \simeq \mathbb{C}^8$ $\quad |5\rangle^3 = |101\rangle = \begin{pmatrix}0\\0\\0\\0\\0\\1\\0\\0\end{pmatrix}$

In ${}^1\mathbb{H}$, let $\begin{cases} |\varphi_1\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} \\ |\varphi_2\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} \end{cases}$

$$\Rightarrow \quad |\varphi_1 \otimes \varphi_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} \otimes \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1\\-1\\1\\-1\end{pmatrix}$$

$$\langle \varphi_1 \otimes \varphi_2| = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix} \otimes \frac{1}{\sqrt{2}}\begin{pmatrix}1 & -1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1 & -1 & 1 & -1\end{pmatrix}$$

2. Bell basis : basis of $\mathcal{H}^{\otimes 2}$.

$\{ |\Phi^{\pm}\rangle := \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle),\ |\Psi^{\pm}\rangle := \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle) \}$.

$\rightarrow$ orthonormal

$$\Big(\langle \Phi^{+}|\Phi^{+}\rangle = \frac{1}{2}\langle 00+11|00+11\rangle = \frac{1}{2}(1+0+0+1) = 1$$

$$\langle \Phi^{+}|\Phi^{-}\rangle = \frac{1}{2}\langle 00+11|00-11\rangle = \frac{1}{2}(1+0+0-1) = 0$$

$\vdots$

Similarly $\langle \Phi^{-}|\Phi^{-}\rangle = \langle \Psi^{\pm}|\Psi^{\pm}\rangle = 1, \cdots \Big)$

(Computational B : sub-sys. of each elt is pure state.
Bell B : " X pure state.

ex) $\mathcal{H}^{2\otimes}$ (Comput B ; $|00\rangle$ $|01\rangle$ $|10\rangle$ $|11\rangle$
Bell B ; $|\Phi^{+}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

3.3. States & Observables of composite systems.

Postulate 7

The Hilbert space of a composite sys. that consists of $\mathbb{H}^A$ and $\mathbb{H}^B$ & $\mathbb{H}^C$ is $\mathbb{H}^A \otimes \mathbb{H}^B \otimes \mathbb{H}^C$

1. State

By Postulate 5, it is described by ρ on $\mathbb{H}^A \otimes \mathbb{H}^B$.

$$\rho = \sum_{j \in I} p_j |\Psi_j\rangle\langle\Psi_j|, \quad \text{where } \{|\Psi_j\rangle \mid j \in I\} : \text{ONB of } \mathbb{H}^A \otimes \mathbb{H}^B,$$

$$p_j \in [0,1], \quad \sum_I p_j = 1.$$

* Convention : associate person – subsys.

Alice controls subsystem A
Bob controls sub-system B

2. Observables

let
M^A : operator on $\mathbb{H}^A$
M^B : operator on $\mathbb{H}^B$

$\Rightarrow$ Construct $\underline{M^A \otimes M^B}$: $\mathbb{H}^A \otimes \mathbb{H}^B \longrightarrow \mathbb{H}^A \otimes \mathbb{H}^B$.

$$|\varphi \otimes \psi\rangle \longmapsto M^A(|\varphi\rangle) \otimes M^B(|\psi\rangle)$$

$$\underset{\sum \Psi_{ab} |e_a \otimes f_b\rangle}{\overset{\shortparallel}{|\Psi\rangle}} \longmapsto \sum_{a,b} \Psi_{ab} M^A(|\varphi\rangle) \otimes M^B(|\psi\rangle).$$

$\rightarrow$ M^A, M^B : self-adj, $\Rightarrow$ $M^A \otimes M^B$: self-adj.

$\left(\because \right.$ For any $|\varphi_1\rangle, |\varphi_2\rangle \in \mathbb{H}^A$, $|\psi_1\rangle, |\psi_2\rangle \in \mathbb{H}^B$,

$$\langle (M^A \otimes M^B)^* \varphi_1 \otimes \psi_1 \mid \varphi_2 \otimes \psi_2 \rangle$$

$$= \langle \varphi_1 \otimes \psi_1 \mid (M^A \otimes M^B)(\varphi_2 \otimes \psi_2) \rangle$$

$$= \langle \varphi_1 \otimes \psi_1 \mid M^A(\varphi_2) \otimes M^B(\psi_2) \rangle$$

$$= \langle \varphi_1 \mid M^A(\varphi_2) \rangle \langle \psi_1 \mid M^B(\psi_2) \rangle$$

$$= \langle (M^A)^* \varphi_1 \mid \varphi_2 \rangle \langle (M^B)^* \psi_1 \mid \psi_2 \rangle$$

$$= \langle (M^A)^* \varphi_1 \otimes (M^B)^* \psi_1 \mid \varphi_2 \otimes \psi_2 \rangle$$

$$= \langle M^A \varphi_1 \otimes M^B \psi_1 \mid \varphi_2 \otimes \psi_2 \rangle$$

$$= \langle (M^A \otimes M^B)(\varphi_1 \otimes \psi_1) \mid \varphi_2 \otimes \psi_2 \rangle$$

$\rightarrow$ matrix repn of $M^A \otimes M^B$?

suppose for $\{|e_a\rangle\}$, $\{|f_b\rangle\}$: basis of $\mathbb{H}^A$, $\mathbb{H}^B$ resp. w. dim n_A, n_B resp.

$$M^A = \begin{pmatrix} M^A_{11} & & \\ & \ddots & \\ & & M^A_{n_A n_A} \end{pmatrix}, \quad M^B = \begin{pmatrix} M^B_{11} & & \\ & \ddots & \\ & & M^B_{n_B n_B} \end{pmatrix}$$

Then w. basis $\{|e_a \otimes f_b\rangle\}$ of $\mathcal{H}^A \otimes \mathcal{H}^B$, $(\dim = n = n_A n_B)$

$$M^A \otimes M^B = \sum_{a,a'=1}^{n_A} \sum_{b,b'=1}^{n_B} |e_a \otimes f_b\rangle\langle e_a \otimes f_b| (M^A \otimes M^B) |e_{a'} \otimes f_{b'}\rangle\langle e_{a'} \otimes f_{b'}|$$

$$= \sum\sum |e_a \otimes f_b\rangle\langle e_a \otimes f_b| M^A e_{a'} \otimes M^B f_{b'}\rangle\langle e_{a'} \otimes f_{b'}|$$

$$= \sum\sum |e_a \otimes f_b\rangle \underbrace{\langle e_a | M^A e_{a'}\rangle}_{M^A_{aa'}} \underbrace{\langle f_b | M^B f_{b'}\rangle}_{M^B_{bb'}} \langle e_{a'} \otimes f_{b'}|$$

$$= \sum_{a,a'=1}^{n_A} \sum_{b,b'=1}^{n_B} \underbrace{M^A_{aa'} M^B_{bb'} |e_a \otimes f_b\rangle\langle e_{a'} \otimes f_{b'}|}_{\|}$$

$$\begin{array}{cc} & (a'-1)n_B + b' \\ (a-1)n_B + b & \begin{pmatrix} 0 & & 0 \\ & M^A_{aa'} M^B_{bb'} & \\ 0 & & 0 \end{pmatrix} \end{array}$$

$$M^A \otimes M^B = \tag{3.34}$$

$$\begin{array}{c|c}
 & \begin{array}{ccc|ccc|cc} 1 & \dots & n_B & n_B+1 & \dots & 2n_B & \dots & n_A n_B \end{array} \\
\begin{array}{c} 1 \\ \vdots \\ n_B \\ n_B+1 \\ \vdots \\ 2n_B \\ \vdots \\ n_A n_B \end{array} &
\left(\begin{array}{ccc|ccc|cc}
M^A_{11}M^B_{11} & \dots & M^A_{11}M^B_{1n_B} & M^A_{12}M^B_{11} & \dots & M^A_{12}M^B_{1n_B} & \dots & M^A_{1n_A}M^B_{1n_B} \\
\vdots & & \vdots & \vdots & & \vdots & & \vdots \\
M^A_{11}M^B_{n_B1} & \dots & M^A_{11}M^B_{n_Bn_B} & M^A_{12}M^B_{n_B1} & \dots & M^A_{12}M^B_{n_Bn_B} & \dots & M^A_{1n_A}M^B_{n_Bn_B} \\
\hline
M^A_{21}M^B_{11} & \dots & M^A_{21}M^B_{1n_B} & M^A_{22}M^B_{11} & \dots & M^A_{22}M^B_{1n_B} & \dots & M^A_{2n_A}M^B_{1n_B} \\
\vdots & & \vdots & \vdots & & \vdots & & \vdots \\
M^A_{21}M^B_{n_B1} & \dots & M^A_{21}M^B_{n_Bn_B} & M^A_{22}M^B_{n_B1} & \dots & M^A_{22}M^B_{n_Bn_B} & \dots & M^A_{2n_A}M^B_{n_Bn_B} \\
\hline
\vdots & & \vdots & \vdots & & \vdots & & \vdots \\
M^A_{n_A1}M^B_{n_B1} & \dots & M^A_{n_A1}M^B_{n_Bn_B} & M^A_{n_A2}M^B_{n_B1} & \dots & M^A_{n_A2}M^B_{n_Bn_B} & \dots & M^A_{n_An_A}M^B_{n_Bn_B}
\end{array}\right),
\end{array}$$

$$
= \left(\begin{array}{c|c|c}
M^A_{11} M^B & \cdots & M^A_{1n_A} M^B \\
\hline
\vdots & & \vdots \\
\hline
M^A_{n_A 1} M^B & & M^A_{n_A n_A} M^B
\end{array}\right),
$$

lem 3.13

let $|\varphi_1\rangle, |\varphi_2\rangle \in \mathbb{H}^A$, $|\psi_1\rangle, |\psi_2\rangle \in \mathbb{H}^B$

$\Rightarrow |\varphi_1 \otimes \psi_1\rangle\langle\varphi_2 \otimes \psi_2| = |\varphi_1\rangle\langle\varphi_2| \otimes |\psi_1\rangle\langle\psi_2|$

Pf) For any $|a_1\rangle, |a_2\rangle \in \mathbb{H}^A$, $|b_1\rangle, |b_2\rangle \in \mathbb{H}^B$,

$$\underbrace{\langle a_1 \otimes b_1 | \left(|\varphi_1 \otimes \psi_1\rangle\right.}\underbrace{\left.\langle\psi_2 \otimes \psi_2|\right) |a_2 \otimes b_2\rangle}$$

$$= \langle a_1|\varphi_1\rangle\langle b_1|\psi_1\rangle\langle\psi_2|a_2\rangle\langle\psi_2|b_2\rangle$$

$$\langle a_1 \otimes b_1 | \underbrace{\left(|\varphi_1\rangle\langle\varphi_2| \otimes |\psi_1\rangle\langle\psi_2| \right) |a_2 \otimes b_2\rangle}.$$

$$= \langle a_1 \otimes b_1 | \left(|\varphi_1\rangle\underline{\langle\varphi_2|a_2\rangle} \otimes |\psi_1\rangle\underline{\langle\psi_2|b_2\rangle} \right)$$

$$= \langle a_1|\varphi_1\rangle\langle b_1|\psi_1\rangle\langle\varphi_2|a_2\rangle\langle\psi_2|b_2\rangle.$$

ex) ${}^{4}\mathbb{H}^{AB} = {}^{2}\mathbb{H} \otimes {}^{2}\mathbb{H}$.

$$M^A \otimes M^B = \sigma_z^A \otimes \sigma_z^B \quad \left(\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right).$$

Consider Bell basis $\{ |\Phi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle),$

$|\Psi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle) \}$.

$$\Rightarrow (\sigma_z^A \otimes \sigma_z^B)|\Phi^{\pm}\rangle = \frac{1}{\sqrt{2}}\{(\sigma_z \otimes \sigma_z)|00\rangle \pm (\sigma_z \otimes \sigma_z)|11\rangle\}$$

$$= \frac{1}{\sqrt{2}}\{\sigma_z|0\rangle \otimes \sigma_z|0\rangle \pm \sigma_z|1\rangle \otimes \sigma_z|1\rangle\}$$

$$= \frac{1}{\sqrt{2}}\{|0\rangle \otimes |0\rangle \pm (-|1\rangle) \otimes (-|1\rangle)\}$$

$$= \frac{1}{\sqrt{2}}\{|00\rangle \pm |11\rangle\}$$

$$= |\Phi^{\pm}\rangle.$$

$$(\sigma_z \otimes \sigma_z)|\Psi^{\pm}\rangle = -|\Psi^{\pm}\rangle$$

$$(\sigma_x \otimes \sigma_x)|\Phi^{\pm}\rangle = \pm|\Phi^{\pm}\rangle$$

$$(\sigma_x \otimes \sigma_x)|\Psi^{\pm}\rangle = \pm|\Psi^{\pm}\rangle$$

For basis $\{ |\Phi^+\rangle, |\Phi^-\rangle, |\Psi^+\rangle, |\Psi^-\rangle \}$.

$$\sigma_z \otimes \sigma_z \Big|_{Bell} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\sigma_x \otimes \sigma_x \Big|_{Bell} = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & 1 & \\ & & & -1 \end{pmatrix}$$

$$[\sigma_z \otimes \sigma_z , \sigma_x \otimes \sigma_x] = 0 .$$

Table 3.1 State determination via joint measurement of $\sigma_z \otimes \sigma_z$ and $\sigma_x \otimes \sigma_x$

Measured value of		State after measurement
$\sigma_z \otimes \sigma_z$	$\sigma_x \otimes \sigma_x$	
$+1$	$+1$	$\lvert \Phi^+ \rangle$
$+1$	-1	$\lvert \Phi^- \rangle$
-1	$+1$	$\lvert \Psi^+ \rangle$
-1	-1	$\lvert \Psi^- \rangle$

Carrying out measurements on subsystem $\mathbb{H}^A$

$\rightarrow$ Measurement of $M^A \otimes \mathbb{1}^B$ of composite sys. $\mathbb{H}^A \otimes \mathbb{H}^B$

$|\Psi\rangle \in \mathbb{H}^A \otimes \mathbb{H}^B$.

$\parallel$

$\sum \Psi_{ab} |e_a \otimes f_b\rangle$

$$\Rightarrow \langle M^A \otimes \mathbb{1}^B \rangle_{\Psi} = \langle \Psi | M^A \otimes \mathbb{1}^B | \Psi \rangle$$

$$= \sum \Psi_{ab} \langle \Psi | \underbrace{(M^A \otimes \mathbb{1}^B) |e_a \otimes f_b\rangle}_{\substack{\parallel \\ M^A e_a \otimes f_b}}$$

$$= \sum_{a,b} \sum_{a',b'} \overline{\Psi_{a'b'}} \Psi_{ab} \underbrace{\langle e_{a'} \otimes f_{b'} | M^A e_a \otimes f_b \rangle}_{\substack{\parallel \\ \langle e_{a'} | M^A e_a \rangle \underbrace{\langle f_{b'} | f_b \rangle}_{\substack{\parallel \\ \delta_{bb'}}}}}$$

$$= \sum_{a,a',b} \overline{\Psi_{a'b}} \Psi_{ab} \langle e_{a'} | M^A e_a \rangle$$

Claim) $\langle M^A \rangle_{\rho^A(\Psi)} = \langle M^A \otimes \mathbb{1}^B \rangle_{\Psi}$,

$$\rho^A(\Psi) := \sum_{a,a',b} \overline{\Psi_{a'b}} \Psi_{ab} |e_a\rangle\langle e_{a'}|$$

Claim) $\rho^A(\Psi)$ is density oper.

pf) 1. self-adj

$$\left(\rho^A(\Psi)^* = \rho^A(\Psi) \right)$$

2. postv

$$\Big(\langle \varphi | \rho^A(\Psi) \varphi \rangle = \sum_{a,a',b} \overline{\Psi_{a'b}} \Psi_{ab} \langle \varphi | e_a \rangle \langle e_{a'} | \varphi \rangle$$

$$= \sum_b \left(\sum_a \Psi_{ab} \langle \varphi | e_a \rangle \right) \left(\overline{\sum_{a'} \Psi_{a'b} \langle \varphi | e_{a'} \rangle} \right)$$

$$= \sum_b \left| \sum_a \Psi_{ab} \langle \varphi | e_a \rangle \right|^2 \geq 0 \Big)$$

3. $tr = 1$

$$tr(\rho^A(\Psi)) = \sum_{a',a,b} \overline{\Psi_{a'b}} \Psi_{ab} \underbrace{tr(|e_a\rangle\langle e_{a'}|)}_{\substack{= \sum_c \langle e_c | e_a \rangle \langle e_{a'} | e_c \rangle \\ = \sum_c \delta_{ac} \delta_{a'c}}}$$

$$= \sum_{a,b} \overline{\Psi_{ab}} \Psi_{ab}$$

$$= \| \Psi \|^2 = 1.$$

Pf) $\langle M^A \rangle_{\rho^A(\Psi)} = tr(\rho^A(\Psi) M^A)$

$$= \sum_c \langle e_c | \rho^A(\Psi) M^A e_c \rangle$$

$$= \sum_{a,a',b,c} \overline{\Psi_{a'b}} \Psi_{ab} \underbrace{\langle e_c | e_a \rangle}_{= \delta_{ac}} \langle e_{a'} | M^A e_c \rangle$$

$$= \sum_{a,a',b} \overline{\Psi_{a'b}} \Psi_{ab} \langle e_{a'} | M^A e_a \rangle$$

$$= \langle M^A \otimes \mathbb{1}^B \rangle_\Psi .$$

$$\rho = |\Psi\rangle\langle\Psi| .$$

cf. $\langle M^B \rangle_{\rho^B(\Psi)} = \langle \mathbb{1}^B \otimes M^B \rangle_\Psi$